CS7646

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1 INTRODUCTION

This report compares a rule-based Manual Strategy against a data-driven Random Forest Strategy Learner trading JPM, with a benchmark of buy-and-hold 1000 shares. Both strategies draw on the same three indicators from Project 6. I hypothesize

1. The Manual Strategy outperforms the benchmark in-sample by exploiting mean-reversion.
2. The Strategy Learner beats the Manual Strategy in-sample because it learns from data, but overfits out-of-sample.
3. Raising market impact degrades Strategy Learner returns monotonically.

1.1 Methods

I use three indicators, each with a 14-day lookback. All three generate clear entry thresholds.

1. Relative Strength Index (RSI): Computed as "100 − 100/(1+RS)" where "RS = avg_gain / avg_loss" over a 14-day rolling window. Values below 30 indicate oversold conditions (long signal); values above 70 indicate overbought conditions (short signal).
2. EMA Ratio: "price / EMA(14)", where EMA uses pandas "ewm(span=14, adjust=False)". I interpret it as a mean-reversion signal: ratio above 1.02 means price has stretched above its recent average (short), below 0.98 means price has fallen below (long).
3. Bollinger Bands: "(price − lower) / (upper − lower)", where bands are SMA(14) ± 2*stdev(14). Values below 0 are outside the lower band (long), above 1 are outside the upper band (short).

All three parameters (lookback = 14, Bollinger Band stdev multiplier = 2, RSI thresholds 30/70) are standard technical-analysis defaults and kept identical between Manual and Strategy Learner.

2 DISCUSSION

2.1 Manual Strategy

Each indicator casts a vote in {-1, 0, +1} using the thresholds above. Votes are summed. A sum ≥ +1 targets a long 1000-share position, a sum ≤ -1 targets a short 1000-share position, otherwise no changes. Trades are computed as the first difference of the target series. Entry thresholds are on opposing ends of the spectrum (RSI 30/70 vs. EMA 0.98/1.02) to capture each indicator's extremes. The vote methodology ensures each indicator will contribute to trade signals.

I believe this is effective because the in-sample window (Jan 2008 – Dec 2009) spans the 2008 financial crisis — a regime where mean-reversion dominates. When JPM crashes sharply, all three oversold signals align and produce high-conviction long entries near local bottoms.

In the In-Sample vs Out-of-Sample tests, I use commission = $9.95, impact = 0.005, starting cash = $100,000. Results are below.

<u>Performance Metrics Table:</u>

Metric	Benchmark(In Sample)	Manual(In Sample)	Benchmark(Out of Sample)	Manual(Out of Sample)
Cumulative Return	0.012325	0.280678	−0.083579	−0.123239
Sharpe Ratio	0.130751	0.623133	−0.213837	−0.461031
Std Dev	0.014165	0.009885	0.007078	0.005689
Mean Daily Return	0.000117	0.000388	−0.000095	−0.000165

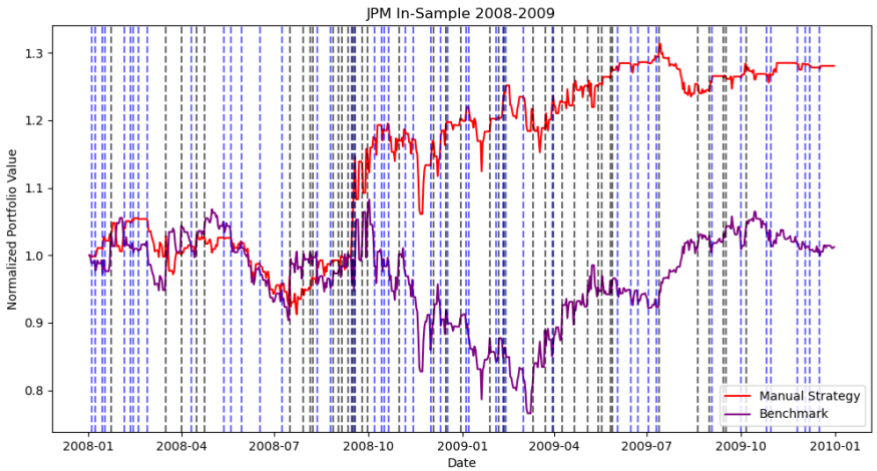

Figure 1— In sample Benchmark and Manual Strategy from 2008 through 2009. Blue lines are long entries, and black are short entries.

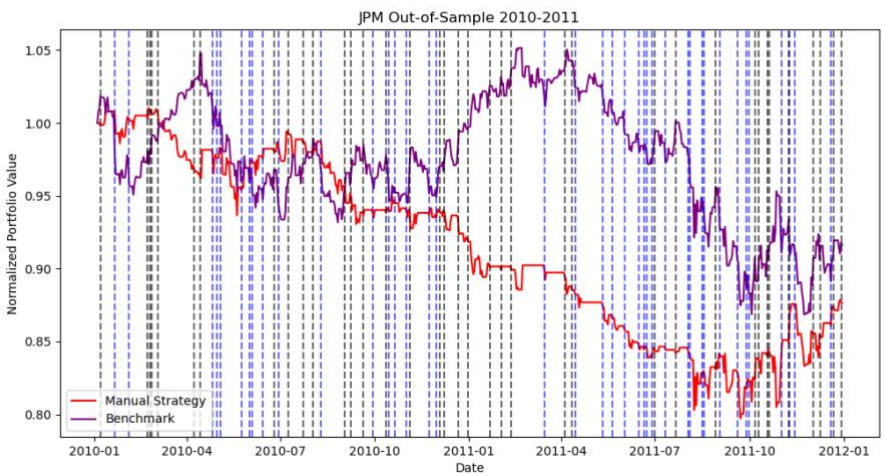

Figure 2— Out of sample Benchmark and Manual Strategy from 2010 through 2011. Blue lines are long entries, and black are short entries.

The Manual Strategy outperforms the benchmark by about 26.8 percentage points of cumulative return and roughly 4.8x Sharpe ratio in-sample. Out-of-sample, it underperforms the benchmark by only about 4.0 percentage points. This reversal reflects the strategy's implicit assumption that mean-reversion dominates.

The in-sample window contains the 2008 crash, when JPM swung from ~$45 to ~$20 and back. Mean-reversion signals produced high-conviction long entries near the lows and exits near the recovery. The out-of-sample window (Jan 2010 – Dec 2011) is dominated by the European sovereign debt crisis, where JPM declined steadily from ~$45 to ~$30 with lower volatility and weaker mean-reverting behavior. Our indicators kept sending oversold signals into a strategy that no longer seemed to work, and the strategy bled capital. I tuned the rule to a regime that doesn't repeat.

2.2 Strategy Learner

We frame the problem as a 3-class classification: for each trading day, predict whether the 10-day forward return will be a "+1"(buy), "-1"(sell), or "0"(hold). Labels are generated by computing "future_ret = prices.shift(-10) / prices – 1" and bucketing: "future_ret > buyHurdle" being +1, "future_ret < -sellHurdle" being -1, otherwise 0. We set "buyHurdle = sellHurdle = 0.02 + impact". The 0.02 floor represents a noise threshold. Smaller moves are likely random and would teach the learner to trade on meaningless signal. Scaling with impact makes the labels impact-aware: a higher impact environment requires bigger expected moves before committing capital.

Features are the same three indicators as Manual Strategy. No standardization is applied: RSI is already bounded in [0,100], Bollinger Band is roughly in [0,1], and EMA ratio is centered near 1.0. Random Trees split on single features at a time and are scale-invariant in the classification setting we use, so normalization provides no benefit.

Hyperparameters for my strategy learner:

- Learner: BagLearner(learner=RTLearner, kwargs={"leaf_size": 5}, bags=20) a Random Forest.
- leaf_size = 5
- bags = 20
- Leaf aggregation: Mode per leaf via np.unique + argmax on training labels.
- Ensemble aggregation: np.sign(sum_across_bags) returns exactly the majority vote over {-1, 0, +1} without floating-point drift.
- lookAhead = 10 days

2.3 Experiment 1: Manual vs Strategy Learner

Hypothesis: The Strategy Learner will beat the Manual Strategy in-sample because it learns feature combinations from data rather than relying on hand-set thresholds, but it will overfit and underperform out-of-sample more severely than the Manual Strategy.

In reality, three indicators with two years of training data are not enough signal for the learner to beat a well-tuned rule, and its added flexibility makes it more fragile when the regime shifts.

In terms of setup, both strategies trade JPM with commission = $9.95, impact = 0.005, sv = $100,000. The Strategy Learner trains on the in-sample window (2008-01-01 to 2009-12-31) and is then evaluated on both the in-sample and out-of-sample (2010-01-01 to 2011-12-31) windows without retraining. The benchmark is buy-and-hold 1000 shares. Random seed = 903011732 set once in testproject.py.

<u>Performance Metrics Results Table:</u>

Metric	Benchmark	Manual	Strategy Learner
In-Sample Cumulative Return	0.012325	0.280678	0.220953
In-Sample Sharpe	0.130751	0.623133	0.451447
Out-of-Sample Cumulative Return	−0.083579	−0.123239	−0.450157
Out-of-Sample Sharpe	−0.213837	−0.461031	−1.303026

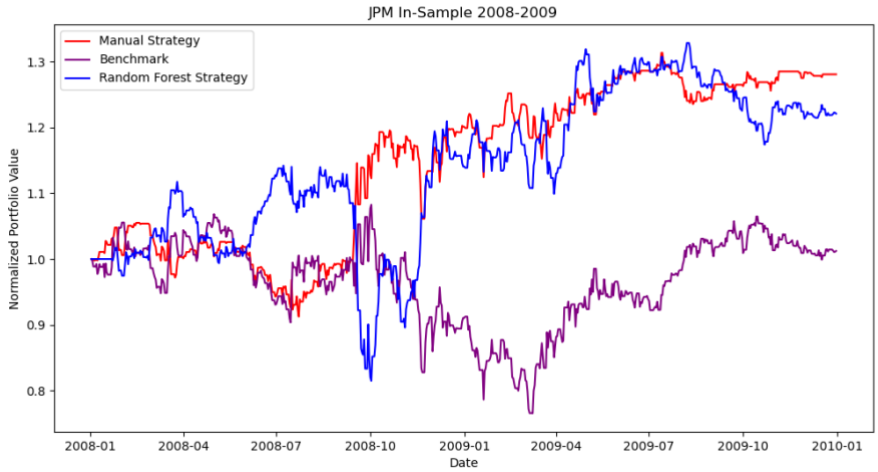

Figure 3 — Manual, Benchmark, and Random Forest Strategies all overlayed in-sample period from 2008 through 2009

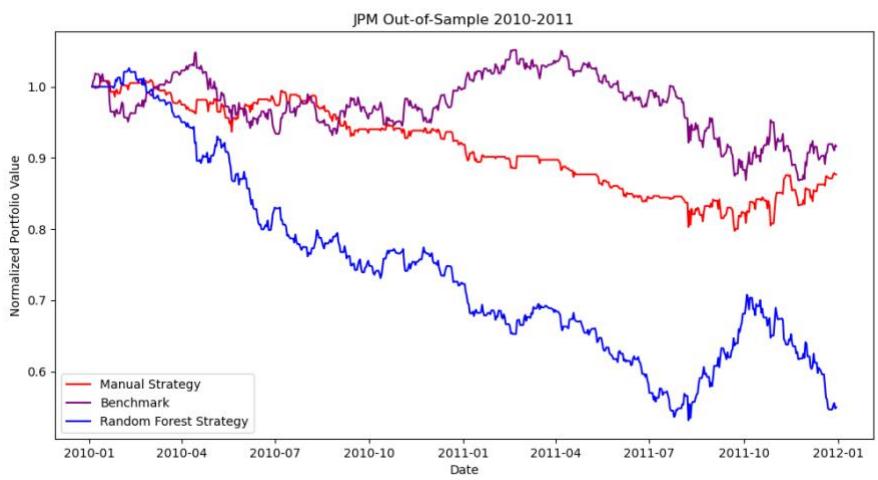

Figure 4 — Manual, Benchmark, and Random Forest Strategies all overlayed out-of-sample period from 2010 through 2011

In-sample, the Strategy Learner returns 22.1% versus the Manual Strategy's 28.1% and the Benchmark's 1.2%. Both beat the benchmark, but the learner does not extract more signal than the hand-built vote aggregator over this window.

Out-of-sample, the learner's cumulative return collapses to -45.0% versus Manual's -12.3% and Benchmark's -8.4%, a loss roughly 5.4x the benchmark's. This confirms that three indicators with about 500 training days are not enough signal for the Random Forest to generalize to a regime it was not trained on.

The behavior of the in-sample results depends on the specific bootstrap samples and split features each bag picks. A different seed would shift the numbers modestly, and on some seeds the learner could plausibly edge above the Manual Strategy. The out-of-sample degradation ordering (Benchmark > Manual > Learner) is expected to persist across seeds because a 20-tree random forest has far more flexibility to memorize 2008 crash patterns than a 6-threshold rule, which makes it the more fragile of the two when the regime shifts.

2.4 Experiment 2: Impact Sensitivity

Hypothesis: as impact rises, the label-generation thresholds "buyHurdle = sellHurdle = 0.02 + impact" grow, fewer training days qualify as confident +1/−1 labels, and the learner's predicted trades respond to a noisier signal. The following occurs:

1. Cumulative return decreases monotonically with impact
2. The number of trades should decrease as hurdles filter marginal labels.

In terms of setup, we followed JPM in-sample only (2008-01-01 to 2009-12-31), commission = $0.00, starting value = $100,000. Four impact values tested: 0.0, 0.005, 0.01, 0.05. For each impact, a fresh StrategyLearner is trained and evaluated. I captured four measurements per metric: cumulative return, number of trades, Sharpe, and Standard Deviation.

<u>Performance Metrics Results Table:</u>

Impact	Cumulative Return	# Trades	Sharpe	Standard Deviation
0	1.118000	126	1.779230	0.009599
0.005	0.322718	147	0.670780	0.010346
0.010	0.144561	147	0.327028	0.013217

0.050	−4.910980	186	0.030316	0.691849

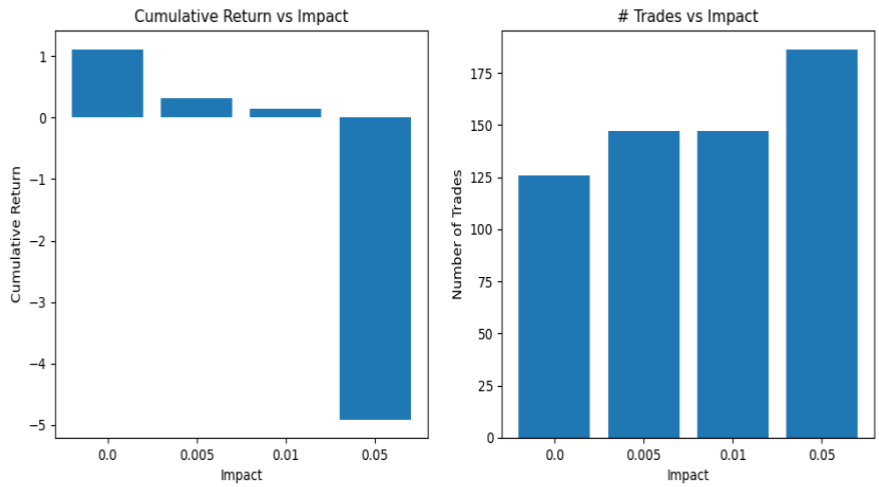

Figure 5— Portfolio value from 2008 through 2009 as impact varies from 0 to 0.05.

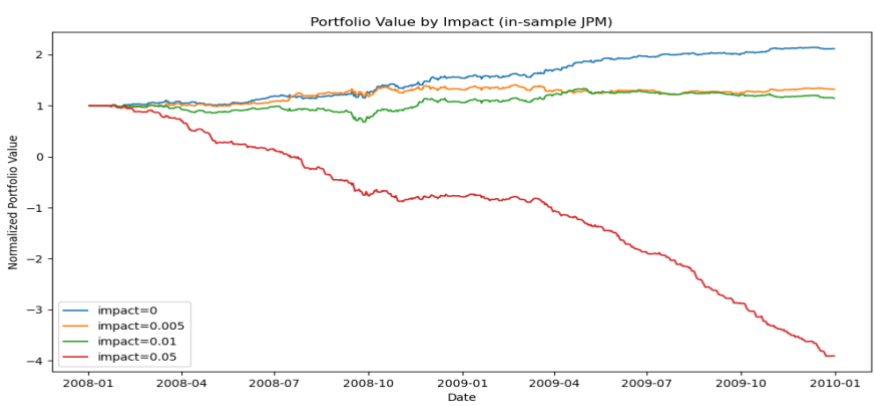

Figure 6— Side-by-side bar charts of cumulative return and # trades vs impact

Cumulative return drops monotonically from +112% at zero impact to -491% at impact = 0.05, confirming the first hypothesis. The Sharpe ratio collapses similarly, and standard deviation explodes at impact = 0.05 as the learner churns. Trade count rises monotonically from 126 to 186 across the four impact levels. Raising the buyHurdle/sellHurdle should intuitively filter out marginal labels

and produce fewer trades, but the opposite happens: at buyHurdle = 0.07 very few days qualify as confident +1 or -1 labels, most training labels collapse to class 0, and the learner ends up with class-0-biased noisy trees that flip predictions frequently on out-of-training-distribution feature combinations. Impact changes behavior, not just returns. Higher impact destabilizes the learner rather than making it more conservative.

The portfolio-value chart makes this vivid: at impact = 0.05 the equity curve diverges from the other three by mid-2008 and spirals into deep negative territory, while the impact = 0 curve rides the in-sample crash-and-recovery cleanly. Any practical deployment should therefore either measure realistic impact from execution data before training, or make impact a tunable hyperparameter rather than a static constant.

3 SUMMARY

Though I initially suspected that the Random Forest Strategy Learner would be able to over-fit and beat-out the manual rule-based hunch I had, this was not validated by the results from this experiment. On JPM, the Random Forest Strategy Learner neither beats the Manual Strategy in-sample nor generalizes out-of-sample. Three indicators with two years of training data are not enough signal to outperform a thoughtfully tuned rule, and the learner's added flexibility makes it more fragile when the regime shifts. Impact is also not a parameter to take lightly; a 10x change in impact turns a profitable learner into a non-working one. Backtests optimize based on past rewards, but those reward signals don't persist into the future in the experiments we have seen.